



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester B.Tech (CBCS) Degree Examinations

May 2024

CV1004 -1 – GREEN BUILDING AND PLANNING
 (CV)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- What do Building Byelaws primarily regulate?**
 - Architectural styles
 - Construction materials
 - Land use and construction practices
 - Interior design elements
- Which organization in India is responsible for formulating and updating the National Building Code?**
 - Ministry of Urban Development
 - National Building Construction Corporation
 - Indian Green Building Council
 - Bureau of Indian Standards
- The purpose of setbacks in urban planning is to**
 - Limit the number of floors in a building
 - Ensure adequate open spaces around a building
 - Regulate the color scheme of a building
 - Determine the building's foundation depth
- What does FAR (Floor Area Ratio) represent in architectural and urban planning terms?**
 - ratio of total built-up area to the total plot area
 - ratio of number of floors to the height of building
 - ratio of open area to the built-up area
 - ratio of building width to its depth in plan layout
- What is the primary purpose of preparing sketch plans in architectural design?**
 - Aesthetic appeal
 - Functional requirements
 - Historical significance
 - Technological advancements
- In architectural drawings, what does the term "site plans" typically refer to?**
 - Drawings of interior spaces
 - Elevations of a building's facade
 - Technical specifications for construction
 - Sketch indicating the layout of outdoor spaces and surroundings
- Which principle of planning focuses on correct alignment and positioning of a building on its site?**
 - Principle of architectural composition
 - Principle of orientation
 - Principle of contrast
 - Principle of unity
- Which drawing technique provides a realistic and detailed representation of a building's exterior?**
 - Site plan
 - Pictorial drawing
 - Perspective and rendering
 - Working drawing
- What does Vaastu Shastra emphasize in the selection of a site?**
 - Only direction
 - Shape and site
 - Direction, shape, and slope
 - Slope and surroundings
- What is the impact of neighboring structures according to Vaastu?**
 - Affects sunlight and ventilation
 - Enhances privacy
 - Determines interior decor
 - No significance
- How does Vaastu advise on the significance of land shape?**
 - Shape is irrelevant
 - Only focus on size
 - Consider symmetry and proportion
 - Prefer irregular shapes for uniqueness

12. In Vaastu Shastra, which direction is commonly regarded as favorable for locating the main entrance of a site?
 A) North B) South
 C) East D) West
13. In Vaastu Shastra, what does the shape of a building influence?
 A) Aesthetic appeal B) Flow of energy
 C) Room temperature D) Structural stability
14. What does Vaastu Shastra suggest about the placement of mirrors in a living room?
 A) Facing a window B) Facing each other
 C) Opposite the main entrance D) No specific guidelines
15. How does Vaastu Shastra advise incorporating water elements in interiors for positive energy?
 A) Place aquariums in the south B) Avoid water elements indoors
 C) Install fountains in the southwest D) Keep water vessels in the northeast
16. What is the significance of using earthy tones in Vaastu-compliant interiors?
 A) Creates a vibrant atmosphere B) Promotes stability and balance
 C) Enhances productivity D) Represents negative energy
17. Which of the following is not a factor influencing architectural development?
 A) Cultural influences B) Historical influences
 C) Fashion trends D) Economic considerations
18. What is the primary characteristic of perceptual structure in architecture?
 A) Aesthetic appeal B) Physical strength
 C) Psychological impact D) User comfort
19. Which architectural composition principle involves the arrangement of elements to create a sense of motion and continuity?
 A) Harmony B) Proportion
 C) Balance D) Rhythm
20. According to architectural principles, what does "balance" aim to achieve in a design?
 A) Visual interest B) Symmetry and equilibrium
 C) Difference and variety D) Completeness and togetherness

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

		Marks	BT*	CO*	PO*
1.	a) Explain the purpose of building byelaws in urban planning and development. Provide two specific examples of building byelaws and their impact on construction.	8	L*2	1	1
	b) Develop a line diagram of a North facing two-bed room residential building with a site area of 20m*14m by assuming suitable dimensions for each component.	8	L3	2	2
2.	a) Explain setbacks and Floor Area Ratio (FAR) in the context of urban planning and building design.	8	L2	1	1
	b) Develop a line diagram of a primary health centre building as per the standards.	8	L3	2	2
3.	a) How do architectural drawings enhance communication among architects, engineers, and builders in the construction process, and what benefits do detailed drawings offer to this collaboration?	8	L2	1	1
	b) Develop a line diagram of a college canteen building as per the standards.	8	L3	2	2

Unit – II

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| 4. | a) | Explain the principles of choosing a site based on Vaastu Shastra, considering factors like direction, shape, and slope. | 8 | L2 | 3 | 1 |
| | b) | How do the principles of Vaastu contribute to the well-being and harmony of occupants in modern buildings? | 8 | L2 | 4 | 1 |
| 5. | a) | Explain the impact of neighboring structures and environmental factors on site selection. | 8 | L2 | 3 | 1 |
| | b) | Show the kitchen layout and interior decor plan based on Vaastu Shastra principles. | 8 | L3 | 4 | 2 |
| 6. | a) | Explain the importance and significance of each element within Vaastu Shastra. | 8 | L2 | 3 | 1 |
| | b) | Explain how Vaastu Shastra guidelines influence the selection of colors, materials, and furnishings in interior design. | 8 | L2 | 4 | 1 |

Unit – III

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| 7. | a) | Explain how the principles of architectural design contribute to the overall aesthetic appeal and functionality of a building. | 8 | L2 | 5 | 1 |
| | b) | Define the term architecture and elaborate on the factors that influence its development. | 8 | L2 | 5 | 1 |
| 8. | a) | Compare various historical periods to illustrate the influence of architectural design. | 8 | L2 | 5 | 1 |
| | b) | Explain the concept of space in architecture, distinguishing between positive and negative space. | 8 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
